

laughter, palilalia, chewing, urge to move, crying without feeling sad, etc

Autonomic. Blushing, mydriasis, change in blood pressure and breathing, apnea, nausea, tachycardia, sweating and the like

Emotional. Anxiety, mirth, feeling of unreality, fear, happiness, anger, sadness, transient acute depression and hypomania, among others

Cognitive. Acalculia, paraphasia, anomic aphasia, recalling memories, going into a trance, out-of-this-world experience, conduction aphasia, hemispatial neglect, reliving past experiences, agraphia, apraxia and so on

Medical applications of electric brain stimulation

Electrical stimulation of the brain is a relatively new technique used to treat chronic pain and tremors associated with Parkinson's disease. It is administered by passing an electrical current through an electrode implanted in the brain. While implantation of electrodes in the brain is used to treat or diagnose several disorders, the term electric brain stimulation is limited to the treatment of tremors and as a pain-management tool for patients suffering from back problems and other chronic injuries and illnesses.

An electric brain stimulation tremor-control device used in treating people suffering from Parkinson's may interfere with or be affected by cardiac pacemakers and other medical equipment. As a result, patients with other implanted medical equipment may not be good candidates for the therapy. Electrical stimulation of the brain, or deep brain stimulation, is effective in treating tremors in up to 88 per cent of Parkinson's disease patients.

An electrode is implanted into the thalamus (part of the brain) of the patient and is attached to an electric pulse generator via an extension wire. The pulse generator is implanted into the patient's pectoral, or

chest area, and the extension wire is tunneled under the skin. The pulse generator sends out intermittent electrical stimulation to the electrode in the thalamus, which inhibits or partially relieves the tremor. The generator can be turned on and off with a magnet, and needs to be replaced every three to five years.

Similar methods have been used to treat chronic pain that responded unfavourably to conventional therapies. A remote transmitter allowed these patients to trigger electric stimulation to relieve their symptoms on an as-needed basis. Patients with failed back syndrome, trigeminal neuropathy (pertaining to the fifth cranial nerve) and peripheral neuropathy fared well for pain control with this treatment, while patients with spinal cord injury and postherpetic neuralgia (pain along the nerves following herpes) did poorly.

Implantation of electrodes into the brain carries risks of hemorrhage, infarction, infection and cerebral edema. These complications could cause irreversible neurological damage. Patients with an implanted electric brain stimulation tremor-control device may experience headaches, disequilibrium (disturbance of the sense of balance), burning or tingling of skin or partial paralysis.

Warning over electrical brain stimulation

Preliminary research on the effects of tDCS has spurred a host of scientists (and amateurs) to explore the possibility of improving cognitive function, increasing reaction times and treating mental illnesses through carefully-applied electrodes.

Research has shown that by delivering electricity to the right part of the brain, we can change the threshold of neurons that transmit information in our brain and, by doing that, we can improve cognitive abilities in different types of psychological functions. The idea is to make the neurons more likely to fire.

Preliminary research suggests that electrical stimulation can improve attention as well as have a positive impact on people with cognitive impairments and depression. Research has also highlighted that electrical brain stimulation could have favourable effects on humans, but some companies are selling such devices online, leading to calls to regulate the technology. If used in the wrong way, these devices could be harmful.

We need to know how long to stimulate, at what time to stimulate and what intensity to use. Electrical stimulation is used in a controlled environment for no more than ten minutes at a time and only on participants who have passed strict medical checks. Scientists are, after all, applying electrodes to the brain, which they say could have some unintended results. For example, different brain regions than those intended might be affected and, in some instances, stimulation could impair rather than improve function if the polarity of the stimulation is reversed.

What is even more worrisome is that people are increasingly making DIY brain stimulation kits themselves. This puts the technology in the realms of clever teenagers.

Another concern is that the science behind these devices is not ready for the commercial market, and companies are jumping on the hype of research that is not quite ready for the world. Any device with medical claims that it is meant to affect the biological function should be appropriately regulated.

Suggestions of increased attention and alleviation of certain medical conditions mean that interest in electrical stimulation is bound to increase, but if the research continues to show promising results, it is clear that electric brain stimulation will need to be treated with some caution. **EFY**